# Uchaaran

## A little practice. A deeper connection.

Uchaaran Helper · Kids Mode for Sundar Gutka

**Team 12 · Khalis Foundation**

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### The problem we are solving

Children need help pronouncing Gurbani and a reason to return to daily practice. Conversations with Khalis focused our design on two needs: engagement that sustains practice, and understanding that goes beyond repeating words.

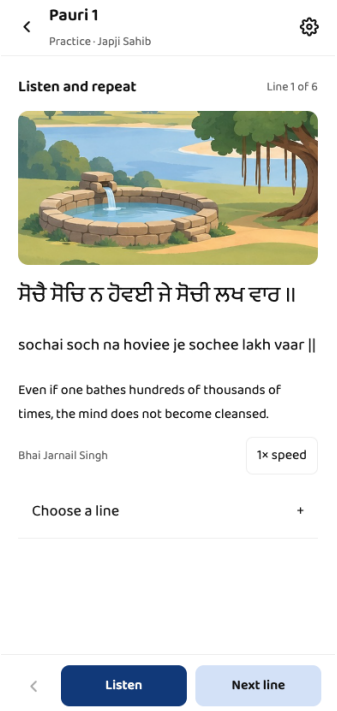
## Two modes. One path toward mastery.

#### 1. Practice at your pace

Listen and repeat with Gurmukhi, Roman-letter pronunciation and English meaning. Use recall mode to practice from memory.

#### 2. Take the pauri challenge

Recite a line and tap Done. Azure returns word-level feedback. Revisit difficult words across 38 pauris; stars and Nanak coins unlock character rewards.




### What changed since midterm

Organizers asked for less pressure and simpler feedback. We added line-by-line scoring, short word prompts, and an ML maintenance guide for Khalis.

### Rooted in the Panth

Built for children, initially grades 6 to 8. Gurmukhi and the ustaad's reference recitation stay central. Familiar landmarks form a memory palace for recall; Sikh-inspired characters and seva stories connect learning with community.

### Tools and implementation

Expo · React Native · TypeScript · Azure-hosted IndicConformer scoring · BaniDB text · reference audio. Codex and Paper supported development and design.

### Private processing, by design

Recordings are processed privately in Azure. Temporary server recordings are deleted after scoring, with no use for public-model training.

Khalis can host the model in its own shared AWS or Azure environment. On-device packaging is another option for fully local processing that keeps recordings on device.

### Ready for handover

MIT source, setup and ML maintenance docs are included. Khalis supplies hosting and credentials; costs depend on usage. An ongoing technical owner remains to be named.

Watch demo ↗ | Try the app ↗ | View code ↗ | Design process ↗